Problem. Each decision step of the cached executor observes a single scalar, the fused retrieval score s_t, and must choose one of $K{+}1$ actions: full inference (action 0) or one of K reuse tiers $a \in \{1, \dots, K\}$ of increasing skip depth $\tau_1 < \cdots < \tau_K$ (in the current system $K = 2$: warm start at $\texttt{start\_t} = 0.3$ and full replay). Unit compute costs are fixed and known,

$$c_a \;=\; \mathrm{s}_1 \;+\; \mathbf{1}[a < K]\,\big(\mathrm{s}_2 + \texttt{start\_t}(a)\cdot \mathrm{s}_3\big), \qquad \texttt{start\_t}(0) = 1, \quad \texttt{start\_t}(K) = 0, \quad (1)$$

so that $c_K < \cdots < c_1 < c_0 = \mathrm{s}_1 + \mathrm{s}_2 + \mathrm{s}_3$, with $\mathrm{s}_1, \mathrm{s}_2, \mathrm{s}_3$ the measured stage latencies (vision encoding, language-model prefix, full flow integration). Every decision pays s_1, which produces the features the retrieval score is computed from; a warm tier resumes the flow at $\texttt{start\_t}(a)$ and pays that fraction of s_3; the deepest tier replays the cached chunk and skips both the prefix and the integration. The online rule class we consider is the incumbent one: *monotone cut vectors*. A rule is a vector $\theta = (\theta_1, \dots, \theta_K)$, $\theta_1 \le \cdots \le \theta_K$ interpreted through "execute the deepest tier whose cut is cleared", $a(s) = \max(\{0\} \cup \{a : s \ge \theta_a\})$. Two ways of choosing θ are compared throughout. The incumbent, which we call the *Grid-Searched Threshold* (GST), sets the cut vector at development-score percentiles and finds operating points by rolling out a grid of them; the alternative, the *Risk-Indexed Threshold* (RIT), computes the cut vector from calibrated risk curves at one tolerance. Both are members of this class at run time and differ only in *how θ is chosen offline*. This note makes that difference the object of study.

Two indexings of one family. GST is *rate indexing*: it sets each θ_a at a percentile of the development score distribution: $\theta_a = \hat{F}_s^{-1}(1 - f_a)$ admits a target fraction f_a of decisions into tier a or deeper. Rates are predictable by construction; *risk* is not. Which vector $(f_1, \dots, f_K)$ lands on the cost–success frontier is unknown a priori, so GST finds operating points by rolling out a K-dimensional grid. RIT is *risk indexing*. Let $Y_a \ge 0$ be the offline-checkable deviation surrogate of the companion note (Eq. 2 there) for tier a, and define per tier the conditional upper quantile

$$q_a(s) \;=\; Q_{1-\alpha}\big(Y_a \mid s_t = s\big), \qquad a = 1, \dots, K, \qquad (2)$$

estimated on calibration rollouts by order-restricted (isotonic) quantile regression under the monotonicity assumption (A1′): q_a nonincreasing in s, nondecreasing in the skip depth. Given one tolerance δ, the cuts are *computed*, not searched:

$$s_a^*(\delta) \;=\; \min\{\, s : \hat{q}_a(s) \le \delta \,\}, \qquad \theta_a = s_a^*(\delta). \qquad (3)$$

The rule executes the deepest admissible tier and falls back to full inference when no tier is admissible (fail-closed, as in the companion note). Online, Eq. **??** is indistinguishable in form and cost from GST: the same comparisons against stored constants.

Equivalence at $K = 1$ (a concession). With a single tier the two indexings coincide as families. Under (A1′) the map $\delta \mapsto s_1^*(\delta)$ is a monotone reparametrization of the cut, so $\{\text{RIT rules}\}_\delta = \{\text{GST rules}\}_\theta = \{\text{monotone rules on } s\}$, and under the monotone-likelihood-ratio premise already stated in the companion note the common family traces the ROC frontier of s (Neyman–Pearson). At $K = 1$, therefore, RIT buys semantics but no new operating points, and percentile inversion is equally automatic. Any claimed contribution must—and here does—begin at $K \ge 2$.

Coupling at $K \ge 2$: the solution path. Consider the budgeted design problem over cut vectors,

$$\max_{\theta}\ \mathrm{SR}(\theta) \qquad \text{s.t.} \qquad \mathbb{E}\big[c_{a(s;\theta)}\big] \le B. \qquad (4)$$

Introduce the surrogate objective in which closed-loop degradation is exchanged for accumulated per-decision surrogate risk,

$$\text{(A2)} \qquad \mathrm{SR}(\theta) \;\approx\; \Phi\Big(\mathbb{E}\big[\, r_{a(s;\theta)}(s) \,\big]\Big), \qquad r_a(s) = \text{a nondecreasing functional of } q_a(s),\ r_0 \equiv 0, \qquad (5)$$

with Φ decreasing. (A2) is an approximation, not a theorem: it ignores sequencing and closed-loop intervention (the composition failure of the companion note) and is adopted because its consequences are checkable. Under (A1′), (A2), and $c_K < \cdots < c_0$, the Lagrangian of Eq. **??** separates across decisions and across adjacent tier boundaries: at an optimum, each boundary θ_a equalizes marginal surrogate risk against marginal compute saving with the *same* multiplier λ,

$$\frac{\partial\, \mathbb{E}[r]}{\partial \theta_a} \;=\; \lambda\, \frac{\partial\, \mathbb{E}[c]}{\partial \theta_a} \qquad (a = 1, \dots, K), \qquad (6)$$

so the set of optimal cut vectors is a *one-parameter solution path* indexed by λ. When the marginal risk at a boundary is governed by the calibrated quantile at the cut—the case in which admission at θ_a adds decisions of surrogate risk $\approx \hat{q}_a(\theta_a)$— the path is traced by a common level: $\hat{q}_a(\theta_a) = \delta(\lambda)$ for all a, which is exactly Eq. **??**. The RIT ladder $\{\, \theta(\delta) \,:\, \delta \geq 0 \,\}$ is thus the plug-in estimate of the solution path of the K-dimensional design problem, and GST's K-dimensional grid search rediscovers that path pointwise. Two consequences are testable: (i) RIT ladder points should track GST's empirical Pareto hull at matched evaluation budget; (ii) at a fixed number n of a-priori arms, the vertical regret of GST's K-dimensional grid against its own hull grows with K, while RIT remains one-dimensional. Neither consequence asserts that RIT *dominates* GST—it cannot, being a subset; the claim is that the path attains the family's frontier without the search.

Budget addressing. Two refinements turn the ladder into a dial. First, the calibration estimate of q_a in the frozen fit is a step function on equal-frequency bins, so every cut snaps to a bin edge and the reachable operating points are coarse; moreover, order restriction pools adjacent bins into plateaus that are data, not artefacts of the bin count (on the libero_10 table one plateau of q_{full} covers a quarter of all decisions). We therefore fit the same two-layer pinball objective on knot values with linear interpolation between equal-frequency knots, add a slope floor $q_a(x_k) - q_a(x_{k+1}) \geq \varepsilon\,(x_{k+1} - x_k)$ with a fixed total ε-budget over the score range (0.02 here) that removes plateaus, and keep the nesting $q_{\mathrm{warm}} \leq q_{\mathrm{full}}$; the cut is then the exact crossing of the fitted curve with δ, and $\delta \mapsto \theta(\delta)$ is continuous and strictly monotone. On a floor segment the fitted gradient is the ε prior, so the position of a cut inside such a segment is budget-addressed rather than risk-addressed; we record that flag per cut and claim no data-supported risk difference inside it. Second, the user-facing dial is the compute budget: the predicted inference ratio $\widehat{\mathrm{IR}}(\delta) = \sum_t c_{a(s_t;\theta(\delta))}/(Nc_0)$ over the shadow-executed calibration rows is nonincreasing in δ and outcome-blind, so a target ratio is inverted to δ by bisection and then to the cut vector. The ladder is thus one path that can be indexed by risk or by budget, whereas the grid needs a K-dimensional search for either. On a finite table $\widehat{\mathrm{IR}}$ is a step function, so the inversion returns the nearest attainable point and records the gap; closed-loop deployment and the production gate move the realized ratio, which we report against the prediction rather than correct.

Per-task and cross-library transfer. GST conditions on nothing: a percentile vector tuned on one task mixture or one library encodes that mixture's score distribution, and the GST's own documentation records that carrying a threshold across libraries is a known failure mode. RIT recomputes Eq. **??** from the calibrated curves: per-task (Mondrian) cuts $s^*_{a,t}(\delta)$ follow from per-task quantile curves at the same δ when sample sizes permit, and a library change re-runs calibration rather than a K-dimensional (per task, $K \times T$-dimensional) grid. The empirical finding that task-conditional selection improves every family by a comparable margin is consistent with this reading: conditioning is a property of the information set, not of any single rule family; what RIT changes is the *cost of exploiting it*.

What is not claimed. No statement here is a closed-loop guarantee: deployment intervenes on the visitation distribution, and Eq. **??** is a surrogate adopted for its checkable consequences (the companion note's scope paragraph applies verbatim). The path property is relative to rules measurable in s alone; richer information sets—the disagreement statistic v, task identity used online, stateful gates—can move the frontier, and whether they do is an empirical question that our measurements answer in the negative for v (no measurable increment at matched density) and in the positive, equally for all families, for task identity. Finally, the estimated path can miss the frontier where calibration data are sparse; matched-density evaluation, not construction, is what supports the tracking claim.

Experimental consequences. Four readouts follow. (i) *Matched-density frontier comparison*: the RIT ladder against the dense GST grid on the same evaluation set, library, and cost accounting; the tracking claim predicts hull agreement within estimation noise. (ii) *Tuning budget*: frontier quality (hull area or vertical regret) as a function of the number of a-priori arms for each method; the prediction is a widening gap as arms are removed, and a qualitative jump when K grows. (iii) *Risk calibration*: realized per-decision surrogate risk of admitted steps against the declared δ — the plot only RIT can draw in advance. (iv) *System composition*: the full deployed stack adds the production hysteresis gate in front of either method; the ceiling-equality reading predicts that gate plus RIT matches gate plus GST at matched freedom, measured on the same chain. A three-tier instance ($K = 3$, second warm depth unpinned from $\texttt{start\_t} = 0.3$ after its unit cost is frozen) turns consequence (ii) into the discriminating experiment: budget-matched 3-dimensional GST grid against the unchanged one-dimensional RIT ladder.